Zulaikha Zakiullah

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Summary of Qualifications

- Embedded systems and firmware engineer with hands-on experience developing radar-integrated, safety-critical products using C on STM32 and ARM Cortex-M microcontrollers
- Strong background in embedded software development, hardware–software integration, and real-time data acquisition, with expertise in signal processing using MATLAB, Python, and C
- Proven experience designing, testing, and validating custom hardware and embedded systems across traffic safety, medical monitoring, and rehabilitation applications
- Proficient in low-level MCU development and communication protocols (SPI, I2C, UART, CAN, Bluetooth, Ethernet), including peripheral control, timing, and debugging
- Effective cross-functional collaborator, translating complex technical systems into reliable products and user-friendly tools for engineers, researchers, and non-technical stakeholders

Education

University of Waterloo **Sep 2019 – Apr 2024**
BASc in Electrical Engineering *Waterloo, ON*

Technical Skills

Programming Languages: C/C++, C#, Python, MATLAB, Verilog, Java, JavaScript, TypeScript
Embedded Systems & Firmware Development: STM32, Microchip, ESP32, FPGA (XEM7310), ARM Cortex-M
Hardware Design & PCB Layout: KiCad, Altium Designer, Schematic Design, PCB Layout, Signal Integrity
Communication Protocols: SPI, I2C, UART, Bluetooth, J1939 (CAN), Ethernet (TCP/IP), MQTT
Software Tools & IDEs: PyQt, Matplotlib, NumPy, MATLAB Simulink, Visual Studio Code, Eclipse
Testing & Validation: Oscilloscope, Multimeter, Automated Testing (Selenium, Jenkins), Unit Testing (JUnit, Python)

Relevant Experience

A.U.G. Signals Ltd. **Oct 2025 – Present**
Embedded Systems Designer *York, ON*

- Designed and implemented embedded firmware in C for STM32 microcontrollers supporting radar-integrated traffic safety systems
- Developed low-level MCU drivers and real-time functionality for radar data acquisition, timing, and peripheral control
- Implemented radar signal processing algorithms using MATLAB and Python, with production deployment in C
- Performed hardware–software integration, debugging, and validation to ensure reliable performance in safety-critical radar systems
- Designed, developed, and maintained C# applications and software solutions, implementing object-oriented programming principles, debugging complex issues, and optimizing application performance
- Integrated and managed MQTT-based communication for IoT and distributed systems, configuring publish/subscribe messaging, troubleshooting connectivity, and ensuring reliable real-time data exchange

Waterloo Silicon Bioelectronics Laboratory **Sep 2023 – Apr 2024**
FPGA & Embedded Firmware Engineer *Waterloo, ON*

- Led the electrical and hardware testing of the team's custom-designed integrated circuit (IC) used for noninvasive continuous glucose and ketone monitoring
- Designed programs using Verilog and Python to control the chip's operating mode via computer and FPGA (XEM7310)
- Performed various checks using electrical equipment, such as an oscilloscope, to validate the 3 main circuit blocks of the chip: the potentiostat, digital-to-analog converter (DAC), analog-to-digital converter (ADC)
- Developed sampling algorithms using Python and Matplotlib to process and visualize data sent from the on-chip ADC
- Assisted in conducting various electrochemical tests on chip using multiple techniques, such as cyclic voltammetry and chronoamperometry, to confirm chip behaviour by comparing live results to simulations

Onsemi **Jan 2023 – Apr 2023**
Firmware & Hardware Validation Engineer *Waterloo, ON*

- Worked on feature support and hardware validation of the company's RSL15, an ultra-low power wireless microcontroller unit (MCU) designed for connecting smart devices in industrial and medical applications, such as hearing aids
- Automated multiple previously manual tests of RSL15 using Python and C, speeding up overall testing procedure
- Designed and tested a proof of concept temperature control module using Microchip microcontroller and C, implementing UART and I2C for communication

- Validated schematics of new module presented by team and suggested changes to make design more reliable

St. Michael's Hospital (Department of Respirology) **Aug 2021 – Aug 2022**
Full Stack Developer *Toronto, ON*

- Collaborated with developers, healthcare practitioners, and researchers to ensure technical and medical needs were met for a clinical decision support system for asthma, ensuring the system recommended the correct medication based on different patients' asthma conditions, and making improvements to the system using JavaScript (see easthma.ca)
- Designed and deployed website for asthma patients and healthcare providers (see asthmadecisionaid.com)
- Created automation software using Python and Selenium to run end-to-end and integration tests of the electronic asthma system, saving over 100 hours of time for the research team
- Built tools with intuitive UI using Python and PyQt to allow physicians and clinical researchers to make code changes to the system without requiring technical knowledge
- Developed software using Python and NumPy to intake large volumes of data relating to the electronic asthma system and process them into readable metrics, allowing the team to view statistics such as the number of patients and physicians using what parts of the system in a given time period, thereby showing what aspects are most effective

Ford Motor Company **Jan 2021 – Apr 2021**
Software Developer *Waterloo, ON*

- Worked under the product development team to push key features and perform initial testing of the infotainment systems slated for the newest Ford models
- Integrated runtime resource overlay packages in Android OS for infotainment system to load specific app restrictions depending on its location, to ensure all vehicles adhere to driving standards set per country
- Developed an API using Java to enable or disable controls on infotainment system based on vehicle's geographic location
- Wrote additional unit tests using Java and JUnit for infotainment system test suite to increase code coverage to 90%+

Ford Motor Company **May 2020 – Aug 2020**
Test Automation Developer *Waterloo, ON*

- Worked under the test automation team to ensure that newly committed software and firmware features passed all test suites in Jenkins before integrating with the production version of code, and features that failed tests were reported to the appropriate development teams
- Utilized DSL in Jenkins to automatically email specific team members when a build fails multiple consecutive times
- Developed an automated service to run monthly using Python to delete all unused workspaces in Jenkins machines, speeding up the testing pipeline
- Created an API used across the testing suite using Python and the façade design pattern to abstract away unnecessary implementation details, allowing for easier and faster usage

Relevant Projects

SoleQuest (Capstone Design Project) **Apr 2023 – Mar 2024**
Electrical & Firmware Engineer *Waterloo, ON*

- Worked with a team of five to build a cost-effective smart insole for lower limb rehabilitation, featuring an insole containing sensors to measure plantar distribution and a custom mobile application to track the user's rehabilitation progress as well as receive the insole data via Bluetooth in real time and display the data as a visual for the user
- Led the design of schematics and layout of printed circuit board (PCB) for insole using KiCad, with features including: SWD interface for microcontroller, 3V regulated battery supply, and RF antenna for Bluetooth communication
- Developed firmware for STM32WB microcontroller using C, configuring features including 12-bit ADC conversion via DMA to process pressure sensor data, SPI communication, and Bluetooth communication to send data to a custom mobile application (also developed by our team)
- Wrote custom SPI interface using C to enable communication between microcontroller and ICM-20948 IMU using information from datasheets, and processed IMU data using a Madgwick filter

Certifications

Standard CPR and First Aid/AED Level C | *Canadian Red Cross* **Aug 2023 – Aug 2026**

- Certificate number: 104377459

Diplôme d'études en langue française (DELF) Level B2 | *France Education International* **Jun 2019**

- Received for passing the DELF B2 examination, demonstrating a degree of independence that allows one to construct arguments to defend their opinion, explain their viewpoint, and negotiate in French